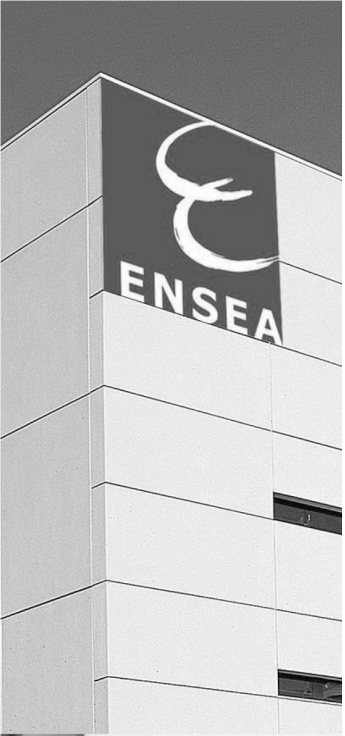
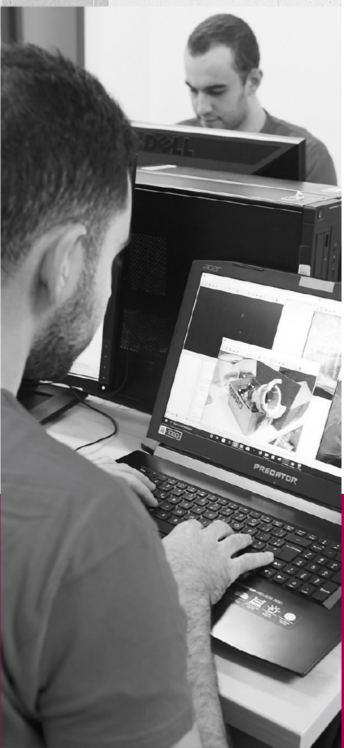




ENSEA

Beyond Engineering

ELECTRONIQUE DE PUISSANCE

DEE_2401

TD

Alexis MARTIN
2026

1 Alimentation d'un microprocesseur

On souhaite alimenter un microprocesseur STM32H4 à l'aide d'une batterie. Le cahier des charges est le suivant:

- Batterie LiPo 2S $V_{in} = 7.4\ V$
- Tension d'alimentation du microprocesseur: $V_{out} = 3.3\ V$
- Ondulation de tension max en entrée du microprocesseur: $\Delta V_S = 50\ mV$
- Courant nominal de sortie: $I_S = 200\ mA$
- Ondulation de courant: 30%

Pour cela, on souhaite utiliser le composant suivant: **LT1959** dont un extrait de la documentation est en annexe. Afin de dimensionner tous les composants, on décide d'utiliser des IAg: Claude (Sonnet 4.6), Copilot (M365 Copilot basé sur GPT-5.5 Automatique) et ChatGPT (GPT 5.5 Instant). Les réponses des IAs ont été générées en juin 2026. Voici le Prompt utilisé:

"Je souhaite utiliser le composant LT1959, pour alimenter un STM32H4 à l'aide d'une batterie. Voici le cahier des charges:

- *Batterie LiPo 2S Vin = 7.4 V*
- *Tension d'alimentation du microprocesseur: Vout = 3.3 V*
- *Ondulation de tension max en entrée du microprocesseur: Δ VS = 50 mV*
- *Courant nominal de sortie: IS = 200 mA*
- *Ondulation de courant: 30%*

Dans un fichier word, détaille moi les calculs pour le dimensionnement des composants passifs. Ajoute une image du schéma électrique. Ajoute une autre image avec la forme du courant dans l'inductance, de la tension sur la broche SW et de la tension de sortie".

A partir des réponses des IAs et de vos connaissances:

1. Représenter le schéma électrique du convertisseur Buck
2. Donner les formes d'ondes de la tension sur la broche SW, du courant dans l'inductance et de la tension de sortie (en supposant que l'ondulation de tension vient principalement de l'ESR du condensateur)
3. Dimensionner les composants passifs du convertisseur Buck

Copilot

1. Cahier des charges

Vin = 7.4 V (LiPo 2S)
Vout = 3.3 V
Iout = 200 mA
Ondulation courant ΔIL = 30%
Ondulation tension ΔVout max = 50 mV

2. Rapport cyclique

D = Vout / Vin = 3.3/7.4 = 0.446

3. Dimensionnement de l inductance

Formule : L = Vout*(Vin - Vout) / (Vin * ΔIL * fs)
ΔIL = 0.3 × 0.2 = 0.060 A
L = 60.95 µH → valeur normalisée : 68 µH
Courant maximal = 0.230 A → choisir > 0.5 A

4. Condensateur de sortie

Formule : Cout = ΔIL / (8 × fs × ΔVout)
Cout = 0.30 µF → choix : 47 µF céramique faible ESR

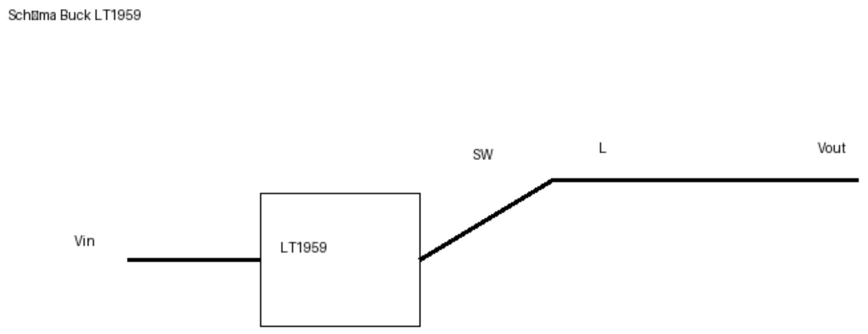
5. Condensateur d entrée

Choix recommandé : 10 µF à 22 µF céramique

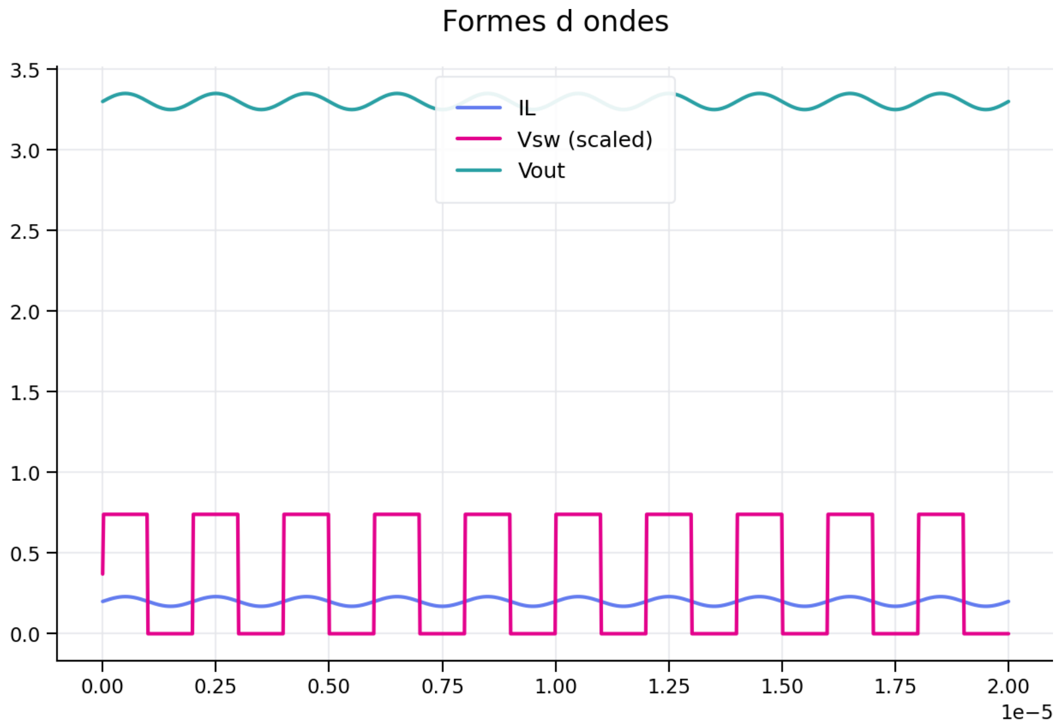
6. Pont diviseur

Vout = Vref (1 + R1/R2)
R2 = 10 kΩ → R1 = 16.83 kΩ → valeur normalisée : 16.9 kΩ

7. Schéma électrique



8. Formes d ondes



Claude

Paramètre	Valeur
Tension d'entrée (batterie LiPo 2S)	VIN = 7,4 V
Tension de sortie	VOUT = 3,3 V
Courant nominal de sortie	IS = 200 mA
Ondulation courant inductance	ΔIL = 30 % de IS
Ondulation tension sortie max	ΔVS = 50 mV
Fréquence de découpage (LT1959)	f = 500 kHz

1. Rapport cyclique (Duty Cycle)

Pour un convertisseur buck idéal en mode continu, le rapport cyclique est :

```
D = VOUT / VIN
D = 3,3 / 7,4
```

D = 0,446 → ton = 0,892 µs / toff = 1,108 µs

Vérification : D = 44,6 % < 50 % → courant limite IP = 4,5 A garanti par le LT1959.

2. Calcul de l'inductance L1

2.1 Ondulation de courant cible

L'ondulation crête-à-crête de courant dans l'inductance est fixée à 30 % du courant de sortie :

```
ΔIL = 0,30 × IS = 0,30 × 200 mA
```

ΔIL = 60 mA

2.2 Valeur minimale de L

La formule issue de la datasheet LT1959 pour le mode continu est :

```
L = (VOUT × (VIN - VOUT)) / (VIN × ΔIL × f)
L = (3,3 × (7,4 - 3,3)) / (7,4 × 0,060 × 500×10³)
L = (3,3 × 4,1) / (7,4 × 30 000)
L = 13,53 / 222 000
```

L_min = 60,9 µH

Valeur normalisée choisie : L1 = 10 µH (série standard, disponible avec Isat > 500 mA).

Remarque : avec L = 10 µH, l'ondulation réelle sera plus faible que la cible de 30 % :

```
ΔIL_réel = (VOUT × (VIN - VOUT)) / (VIN × L × f)
ΔIL_réel = (3,3 × 4,1) / (7,4 × 10×10⁻⁶ × 500×10³)
```

ΔIL_réel ≈ 365 mA → 18 % de IS ✓ (< 30 %)

3. Courants crête et vallée dans l'inductance

```
IL_peak   = IS + ΔIL_réel/2 = 200 + 183  = 383 mA
IL_valley = IS - ΔIL_réel/2 = 200 - 183  =  17 mA
```

Vérification mode continu : IL_valley > 0 ✓

Courant de saturation requis : Isat > IL_peak = 383 mA → inductor standard 10 µH suffisant.

4. Condensateur de sortie C3

4.1 Contrainte ESR (ondulation de tension)

L'ondulation de tension en sortie est principalement déterminée par l'ESR du condensateur :

$$\Delta VS \approx \Delta IL\_réel \times ESR$$

$$ESR\_max = \Delta VS \;/\; \Delta IL\_réel = 50\times10^{-3} \;/\; 0{,}365$$

ESR_max = 136 mΩ

Condensateur choisi : AVX TPS 100 µF / 10 V (tantalum solide, ESR typ. 100 mΩ < 136 mΩ) ✓

4.2 Valeur en capacité

La valeur en µF n'est pas critique. Avec C3 = 100 µF :

$$\text{Ondulation capacitive} = \Delta IL \;/\; (8 \times f \times C3)$$

$$= 0{,}365 \;/\; (8 \times 500\times10^{3} \times 100\times10^{-6})$$

= 0,91 mV (négligeable devant la contribution ESR)

5. Condensateur d'entrée C1

Le courant RMS dans le condensateur d'entrée est :

$$I\_C1\_rms = IS \times \sqrt{(VOUT \times (VIN - VOUT))} \;/\; VIN$$

$$I\_C1\_rms = 0{,}200 \times \sqrt{(3{,}3 \times 4{,}1 \;/\; 7{,}4)}$$

$$I\_C1\_rms = 0{,}200 \times \sqrt{(0{,}183)}$$

I_C1_rms ≈ 85,5 mA

Condensateur choisi : C1 = 100 µF / 10 V, tantalum AVX TPS (Iripple ≥ 0,7 A) ✓

6. Diode de roue libre D1

Courant moyen dans D1 :

$$ID1\_avg = IS \times (VIN - VOUT) \;/\; VIN$$

$$ID1\_avg = 0{,}200 \times (7{,}4 - 3{,}3) \;/\; 7{,}4$$

ID1_avg = 110 mA

Tension inverse maximale = VIN = 7,4 V

Diode choisie : MBRS330T3 (Schottky, 3 A / 30 V, VF typ. 0,5 V @ 3 A) ✓

7. Condensateur BOOST C2

Le condensateur BOOST alimente le driver de l'interrupteur NPN pour le saturer. La valeur minimale est :

$$C2\_min = (IS/50) \;/\; (f \times 0{,}6)$$

$$C2\_min = (0{,}200/50) \;/\; (500\times10^{3} \times 0{,}6)$$

C2_min ≈ 13 nF → C2 choisi = 0,68 µF (valeur datasheet standard) ✓

Diode D2 : 1N914 ou 1N4148, anode connectée à VOUT.

8. Réseau de contre-réaction (R1, R2)

La tension de référence interne est VREF = 1,21 V. Le diviseur résistif R1/R2 fixe VOUT :

$$VOUT = VREF \times (1 + R1/R2)$$

$$\rightarrow\; R1/R2 = (VOUT/VREF) - 1 = (3{,}3/1{,}21) - 1 = 1{,}727$$

Choix de R2 ≤ 2,5 kΩ (recommandation datasheet) : R2 = 2,49 kΩ

$$R1 = R2 \times (VOUT/VREF - 1) = 2490 \times 1{,}727$$

R1 = 4 298 Ω → valeur normalisée : R1 = 4,32 kΩ (E96, 1 %)

Vérification :

```
VOUT = 1,21 × (1 + 4320/2490) = 1,21 × 2,735
```

VOUT = 3,309 V ✓ (erreur < 0,3 %)

9. Condensateur de compensation VC (CC)

La compensation par défaut recommandée par la datasheet pour la majorité des applications est :

```
CC = 1,5 nF  (condensateur seul, RC = 0)
```

Ce réseau crée un pôle à :

```
f_pole = 1 / (2π × ROUT_EA × CC) ≈ 530 Hz
```

Le déphasage de boucle reste > 60° à la fréquence de coupure (≈ 100 kHz), garantissant la stabilité.

10. Bilan des composants

Référence	Valeur / Type	Rôle	Critère de choix
L1	10 µH, Isat > 400 mA	Inductance de filtrage	ΔIL ≈ 18 %
C1	100 µF / 10 V Tant.	Condensateur entrée	Iripple ≥ 86 mA
C2	0,68 µF céramique	Condensateur BOOST	Charge driver NPN
C3	100 µF / 10 V Tant.	Condensateur sortie	ESR ≤ 136 mΩ
CC	1,5 nF céramique	Compensation VC	Stabilité boucle
D1	MBRS330T3 (3A/30V)	Diode roue libre	VF faible, Schottky
D2	1N914 / 1N4148	Diode BOOST	Rapide, faible VF
R1	4,32 kΩ 1 %	Diviseur FB (haut)	Réglage VOUT
R2	2,49 kΩ 1 %	Diviseur FB (bas)	Réglage VOUT

11. Schéma électrique

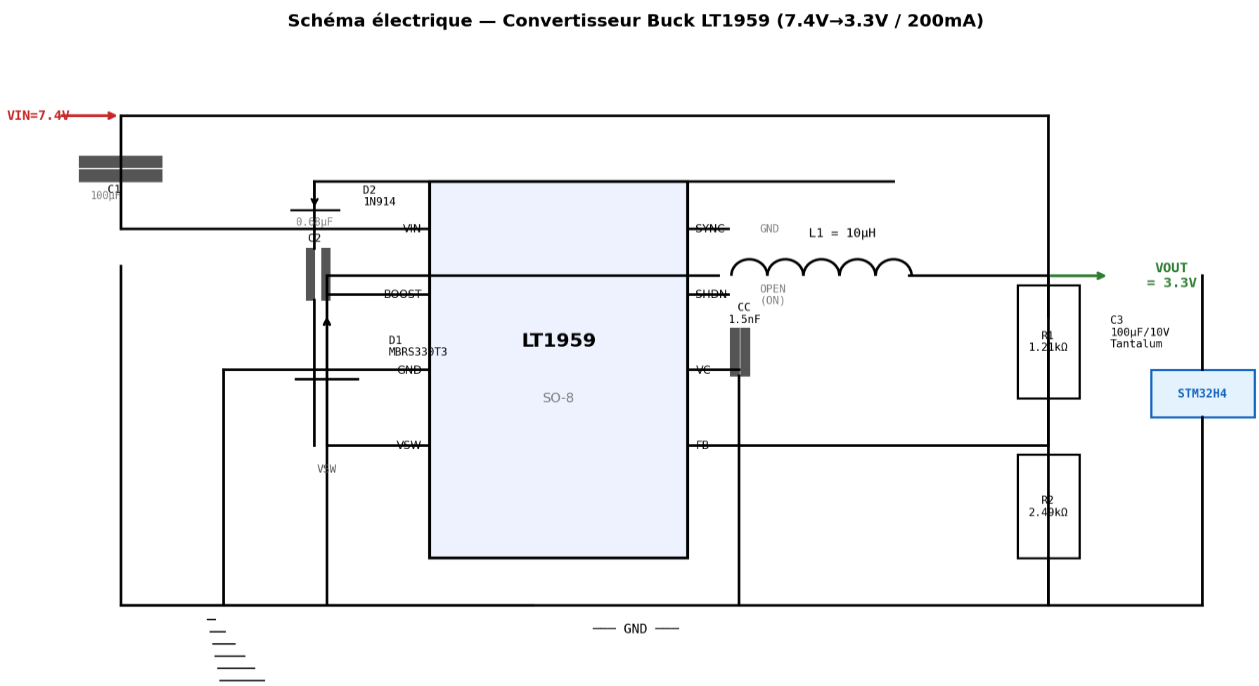


Figure 1 — Schéma de principe du convertisseur Buck LT1959 (7,4 V → 3,3 V / 200 mA)

12. Formes d'ondes simulées

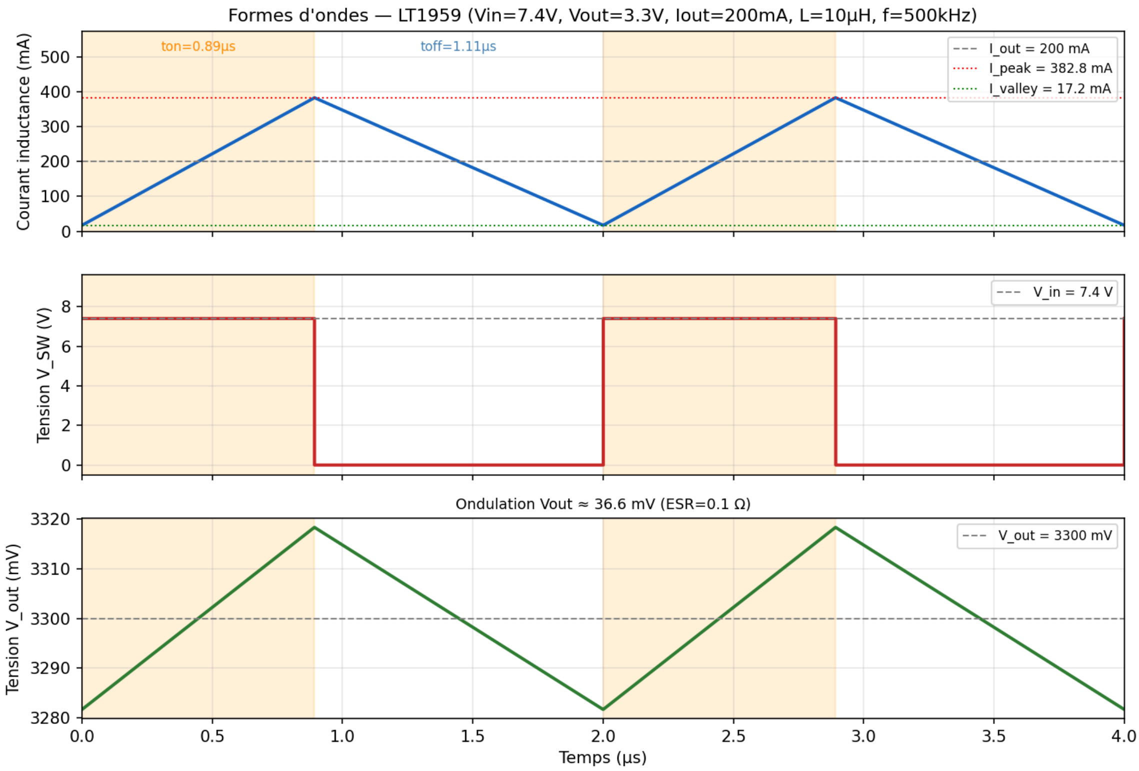


Figure 2 — Courant inductance IL, tension VSW et tension de sortie VOUT sur 2 périodes (f = 500 kHz)

La zone orangée représente le temps de conduction de l'interrupteur (ton ≈ 0,892 µs).

L'ondulation de VOUT est principalement due à l'ESR du condensateur C3 (≈ 36 mV < 50 mV spec) ✓

Référence : Linear Technology — LT1959 Datasheet, Rev. A.

Calculs réalisés pour IS = 200 mA nominal. Vérifier les paramètres thermiques pour des courants supérieurs.

ChatGPT

Objet : dimensionner les composants passifs principaux d'un convertisseur abaisseur LT1959 pour générer 3,3 V / 200 mA à partir d'une batterie LiPo 2S nominale de 7,4 V, avec une ondulation de sortie maximale demandée de 50 mV et une ondulation de courant inductance de 30 %.

1. Données et hypothèses

Grandeur	Valeur	Commentaire
Vin	7,4 V	Tension nominale batterie LiPo 2S
Vout	3,3 V	Alimentation microprocesseur
Is	200 mA	Courant nominal
ΔVS max	50 mV crête-à-crête	Ondulation maximale admise en entrée du STM32
ΔIL	30 % de Is = 60 mA c-à-c	Ondulation de courant inductance
fs	500 kHz	Fréquence fixe du LT1959
VFB	1,21 V	Référence de la broche FB

Remarque importante : les calculs ci-dessous sont réalisés au point nominal Vin = 7,4 V. Pour une conception de série, il faut également vérifier la batterie pleine charge, typiquement 8,4 V pour une LiPo 2S, et la batterie déchargée. Le LT1959 étant largement surdimensionné en courant pour 200 mA, les contraintes dominantes seront plutôt l'ondulation, la stabilité, le choix du condensateur de sortie et le routage.

2. Schéma électrique de principe

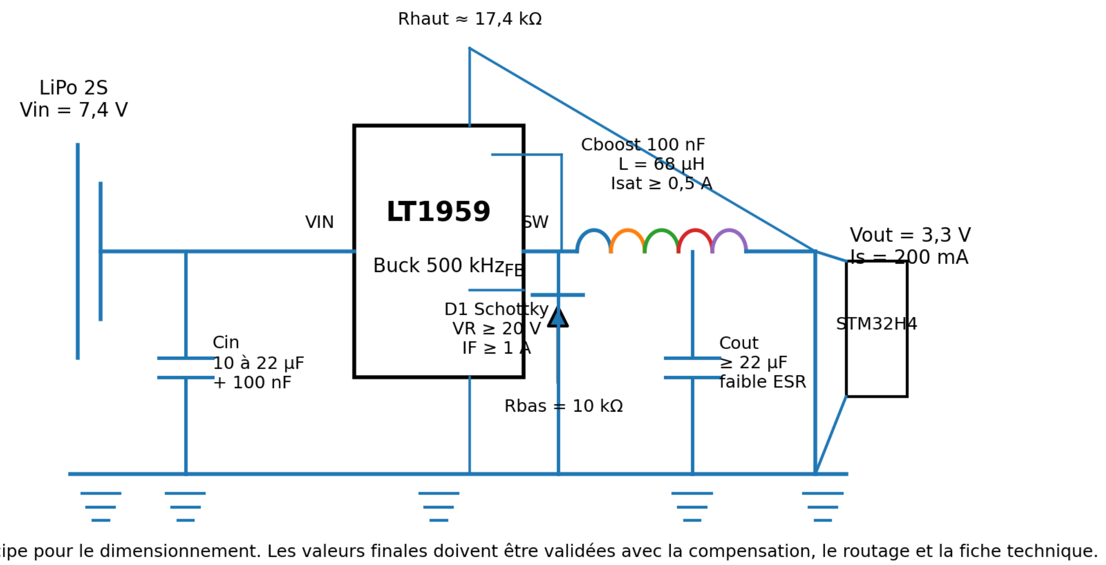


Figure 1 - Schéma de principe du convertisseur buck LT1959 vers 3,3 V.

3. Rapport cyclique

En première approximation, en négligeant les chutes dans l'interrupteur interne et dans la diode de roue libre :

D = Vout / Vin = 3,3 / 7,4 = 0.446, soit 44.6 %.

À 500 kHz, la période de découpage vaut $T = 1 / fs = 2.00$ µs. On obtient $Ton = D.T = 0.89$ µs et $Toff = (1-D).T = 1.11$ µs.

4. Dimensionnement de l'inductance

Pour un buck en conduction continue, l'ondulation crête-à-crête du courant d'inductance est :

$\Delta IL = (Vin - Vout).D / (L.fs)$

Donc $L = (Vin - Vout).D / (fs.\Delta IL) = (7{,}4 - 3{,}3) \times 0.446 / (500\ kHz \times 60\ mA) = 60.9$ µH.

Valeur normalisée conseillée : 68 µH. Avec 68 µH, l'ondulation vaut environ 53.8 mA c-à-c, soit environ 26.9 % du courant nominal.

Courant crête dans l'inductance : $Ipk = Is + \Delta IL/2 = 200\ mA + 30\ mA = 230\ mA$. Courant minimal : $Imin = 170\ mA$. Choisir une inductance avec courant de saturation nettement supérieur, par exemple $Isat \geq 0{,}5$ A, et une résistance série faible pour limiter les pertes à faible tension.

5. Condensateur de sortie

L'ondulation de sortie possède deux contributions principales : la charge/décharge capacitive et la chute ESR du condensateur. La composante capacitive idéale est :

$\Delta Vcap = \Delta IL / (8.fs.Cout)$

En imposant $\Delta Vcap \leq 50$ mV : $Cout \geq \Delta IL / (8.fs.\Delta V) = 60\ mA / (8 \times 500\ kHz \times 50\ mV) = 0.30$ µF.

La contrainte purement capacitive est donc très faible. La contrainte ESR donne : $ESR \leq \Delta V / \Delta IL = 50\ mV / 60\ mA = 0.83\ \Omega$.

Choix pratique recommandé : au minimum 22 µF céramique X7R en sortie, complété si nécessaire par 100 nF proche de la charge. La valeur réelle d'un condensateur céramique baisse avec la polarisation DC : il faut donc vérifier la capacité effective à 3,3 V. Pour un microprocesseur, l'objectif n'est pas seulement l'ondulation à 500 kHz, mais aussi la tenue aux appels de courant rapides.

6. Condensateur d'entrée

Le condensateur d'entrée doit fournir le courant pulsé du convertisseur et rester proche des broches VIN/GND du LT1959. Son courant RMS approximatif vaut :

$Icin,rms \approx Is.sqrt(D.(1-D)) = 0{,}2 \times sqrt(0.446 \times 0.554) = 99$ mA RMS.

Pour viser par exemple 100 mV d'ondulation d'entrée basse fréquence de découpage, une estimation donne $Cin \geq Is.D.(1-D)/(fs.\Delta Vin) = 0.99$ µF. En pratique, retenir 10 à 22 µF faible ESR, avec un 100 nF céramique directement au plus près de VIN et GND.

7. Pont de retour FB

La tension de sortie est fixée par : $Vout = VFB \times (1 + Rhaut/Rbas)$.

En choisissant $Rbas = 10\ k\Omega$: $Rhaut = 10\ k\Omega \times (3{,}3/1{,}21 - 1) = 17.27\ k\Omega$.

Valeur normalisée : $Rhaut = 17{,}4\ k\Omega$ et $Rbas = 10\ k\Omega$, ce qui donne $Vout \approx 3{,}32$ V. Une alternative est $Rhaut = 17{,}2\ k\Omega$ pour obtenir environ 3,29 V.

8. Diode de roue libre et condensateur boost

La diode de roue libre conduit pendant Toff. Son courant moyen vaut Id,moy ≈ Is.(1-D) = 0,2 × 0.554 = 111 mA. Son courant crête est proche du courant crête inductance, soit environ 230 mA au point nominal.

Choisir une diode Schottky rapide avec VRRM ≥ 20 V et IF ≥ 1 A pour disposer d'une marge confortable. La tension inverse doit au minimum supporter Vin, et le routage doit minimiser la boucle LT1959 - diode - condensateur d'entrée.

Le condensateur boost utilisé par la commande interne est typiquement de 100 nF, placé au plus près des broches BOOST et SW, selon le schéma d'application de la fiche technique.

9. Formes d'ondes attendues

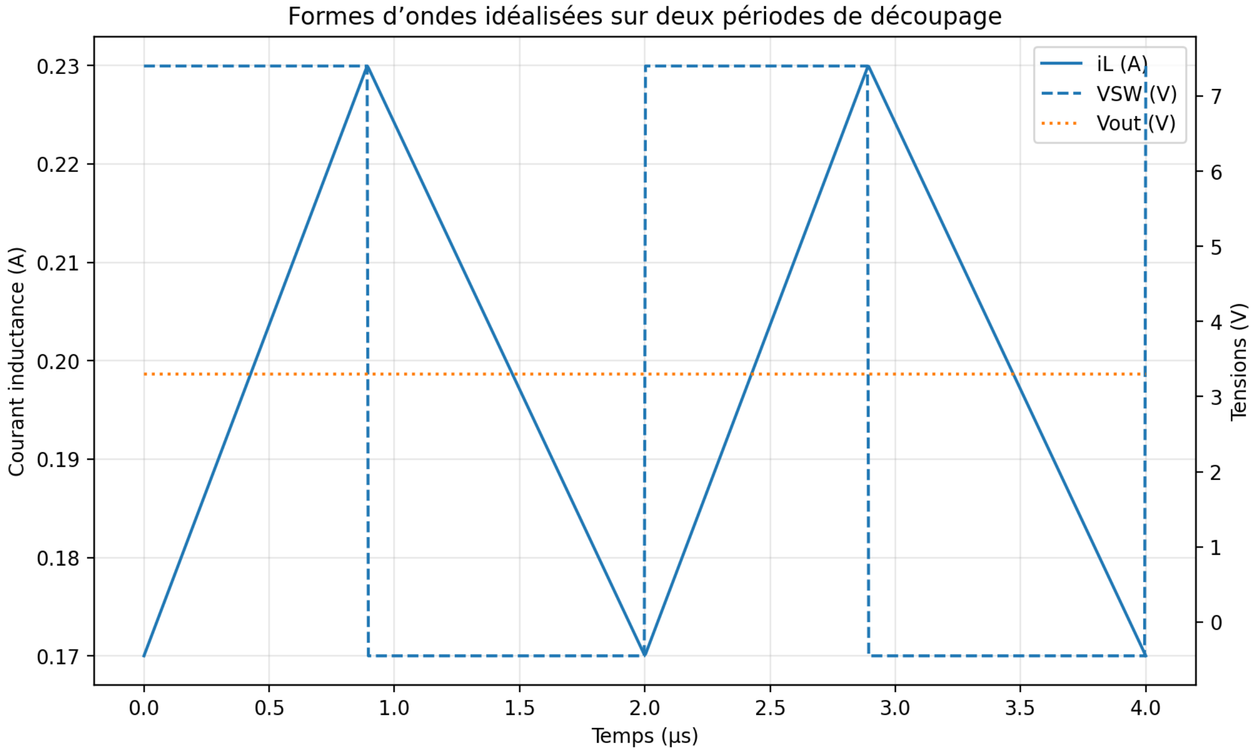


Figure 2 - Courant dans l'inductance, tension SW et ondulation de sortie idéalisés.

Le courant inductance est triangulaire en conduction continue. La broche SW commute entre une valeur proche de Vin pendant Ton et une tension légèrement négative pendant Toff, limitée par la diode de roue libre. La tension de sortie est quasi continue, avec une ondulation résiduelle dépendant de ΔIL, Cout, de l'ESR et du routage.

10. Synthèse des valeurs proposées

Composant	Valeur calculée / contrainte	Choix proposé
L	60,9 µH pour ΔIL = 60 mA	68 µH, Isat ≥ 0,5 A
Cout	≥ 0,30 µF et ESR ≤ 0,83 Ω	22 µF X7R + 100 nF, faible ESR
Cin	Irms ≈ 99 mA ; capacité minimale faible	10 à 22 µF faible ESR + 100 nF proche LT1959
Rhaut/Rbas	Rhaut = 17,27 kΩ si Rbas = 10 kΩ	17,4 kΩ / 10 kΩ
D1	Id,moy ≈ 111 mA ; Icrête ≈ 230 mA	Schottky ≥ 20 V, ≥ 1 A
Cboost	Selon application LT1959	100 nF proche BOOST-SW

Conclusion : au point nominal 7,4 V vers 3,3 V / 200 mA, une inductance de 68 µH et un condensateur de sortie céramique de 22 µF donnent une marge confortable vis-à-vis de l’ondulation de 50 mV. Il faut toutefois valider la stabilité de boucle avec la compensation de la broche VC, les caractéristiques réelles des condensateurs et le routage du circuit imprimé.

Références

Analog Devices, LT1959 product page et fiche technique : régulateur abaisseur 500 kHz, sortie ajustable par FB 1,21 V, interrupteur 4,5 A, entrée minimale 4 V, recommandations de routage VIN/GND/SW/diode.

4.5A, 500kHz Step-Down Switching Regulator

FEATURES

- **Operates with Input as Low as 4V**
- **Output Range Down to 1.21V**
- **Constant 500kHz Switching Frequency**
- **Uses All Surface Mount Components**
- **Inductor Size Reduced to 1.8µH**
- Saturating Switch Design: 0.07Ω
- Shutdown Current: 15µA
- Easily Synchronizable
- Cycle-by-Cycle Current Limiting
- 4.5A Switch
- Current Mode Control

APPLICATIONS

- Portable Computers
- Battery-Powered Systems
- Battery Chargers
- Distributed Power
- 5V to 3.3V Conversion
- 5V to 2.5V Conversion
- 5V to 1.8V Conversion

DESCRIPTION

The LT®1959 is a 500kHz monolithic buck mode switching regulator functionally identical to the LT1506 but optimized for lower output voltage applications. It will operate down to 1.21V output compared to 2.42V for the LT1506. A 4.5A switch is included on the die along with all the necessary oscillator, control and logic circuitry. High switching frequency allows a considerable reduction in the size of external components. The topology is current mode for fast transient response and good loop stability.

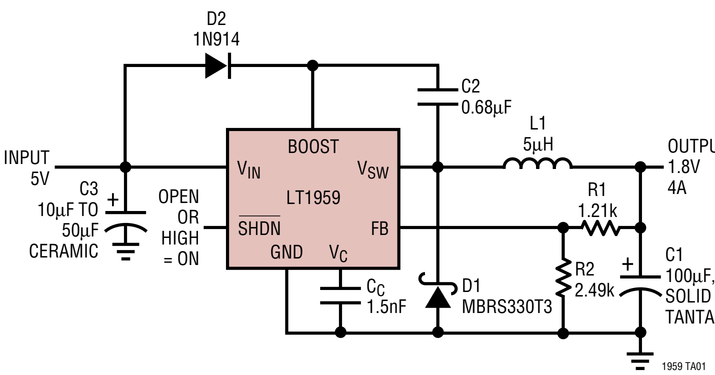
A special high speed bipolar process and new design techniques achieve high efficiency at high switching frequency. Efficiency is maintained over a wide output current range by keeping quiescent supply current to 3.8mA and by utilizing a supply boost capacitor to saturate the power switch.

The LT1959 fits into standard 7-pin DD and fused lead SO-8 packages. Full cycle-by-cycle short-circuit protection and thermal shutdown are provided. Standard surface mount external parts are used, including the inductor and capacitors. There is the optional function of shutdown or synchronization. A shutdown signal reduces supply current to 15µA. Synchronization allows an external logic level signal to increase the internal oscillator from 580kHz to 1MHz.

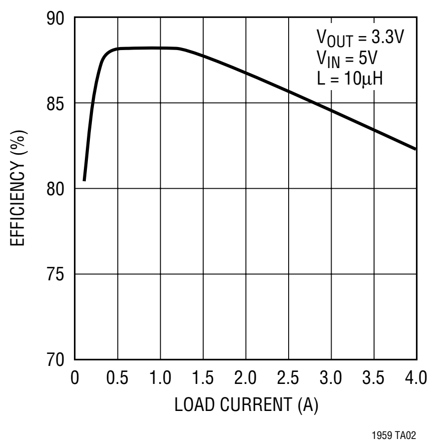
LT, LTC and LT are registered trademarks of Linear Technology Corporation.

TYPICAL APPLICATION

5V to 1.8V Down Converter



Efficiency vs Load Current



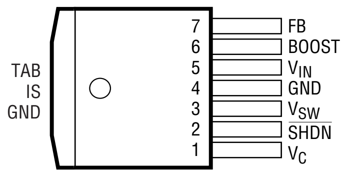
ABSOLUTE MAXIMUM RATINGS (Note 1)

Input Voltage 16V
BOOST Voltage 30V
BOOST Pin Above Input Voltage 15V
$\overline{\text{SHDN}}$ Pin Voltage 7V
FB Pin Voltage 3.5V
FB Pin Current 1mA

SYNC Pin Voltage 7V
Operating Junction Temperature Range
  LT1959C .... 0°C to 125°C
  LT1959I .... –40°C to 125°C
Storage Temperature Range .... –65°C to 150°C
Lead Temperature (Soldering, 10 sec)................. 300°C

PACKAGE/ORDER INFORMATION

FRONT VIEW

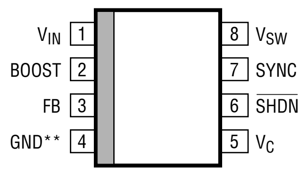


R PACKAGE
7-LEAD PLASTIC DD

T_{JMAX} = 150°C, θ_{JA} = 30°C/W
WITH PACKAGE SOLDERED TO 0.5 SQUARE INCH COPPER AREA OVER BACKSIDE GROUND PLANE OR INTERNAL POWER PLANE. θ_{JA} CAN VARY FROM 20°C/W TO >40°C/W DEPENDING ON MOUNTING TECHNIQUES

ORDER PART NUMBER
LT1959CR
LT1959IR

TOP VIEW



S8 PACKAGE
8-LEAD PLASTIC SO

T_{JMAX} = 150°C, θ_{JA} = 80°C/W
**WITH (FUSED GND) GROUND PIN CONNECTED TO GROUND PLANE OR LARGE LANDS

ORDER PART NUMBER	S8 PART MARKING
LT1959CS8	1959
LT1959IS8	1959I

ELECTRICAL CHARACTERISTICS

The ● denotes the specifications which apply over the full operating temperature range, otherwise specifications are at T_A = 25°C, T_J = 25°C, V_{IN} = 5V, V_C = 1.5V, Boost = V_{IN} + 5V, switch open, unless otherwise noted.

PARAMETER	CONDITIONS		MIN	TYP	MAX	UNITS
Feedback Voltage (Adjustable)	All Conditions	●	1.19	1.21	1.23	V
Reference Voltage Line Regulation	4.3V ≤ V_{IN} ≤ 15V			0.01	0.03	%/V
Feedback Input Bias Current		●	–0.5	0	0.5	µA
Error Amplifier Voltage Gain	(Note 2)		200	400		
Error Amplifier Transconductance	ΔI (V_C) = ±10µA		1500	2000	2700	µMho
		●	1000		3100	µMho
V_C Pin to Switch Current Transconductance				5.3		A/V
Error Amplifier Source Current	V_{FB} = 1.05V	●	140	225	320	µA
Error Amplifier Sink Current	V_{FB} = 1.35V	●	140	225	320	µA
V_C Pin Switching Threshold	Duty Cycle = 0			0.9		V
V_C Pin High Clamp				2.1		V
Switch Current Limit	V_C Open, V_{FB} = 1.05V, DC ≤ 50%	●	4.5	6	8.5	A
Slope Compensation	DC = 80%			0.8		A

LINEAR TECHNOLOGY

ELECTRICAL CHARACTERISTICS

The ● denotes the specifications which apply over the full operating temperature range, otherwise specifications are at T_A = 25°C, T_J = 25°C, V_{IN} = 5V, V_C = 1.5V, Boost = V_{IN} + 5V, switch open, unless otherwise noted.

PARAMETER	CONDITIONS		MIN	TYP	MAX	UNITS
Switch On Resistance (Note 7)	I_{SW} = 4.5A			0.07	0.1	Ω
		●			0.13	Ω
Maximum Switch Duty Cycle	V_{FB} = 1.05V		90	93		%
		●	86	93		%
Switch Frequency	V_C Set to Give 50% Duty Cycle		460	500	540	kHz
		●	440		560	kHz
Switch Frequency Line Regulation	4.3V ≤ V_{IN} ≤ 15V	●		0	0.15	%/V
Frequency Shifting Threshold on FB Pin	Δf = 10kHz	●	0.5	0.7	1.0	V
Minimum Input Voltage (Note 3)		●		4.0	4.3	V
Minimum Boost Voltage (Note 4)	I_{SW} ≤ 4.5A	●		2.3	3.0	V
Boost Current (Note 5)	I_{SW} = 1A	●		20	35	mA
	I_{SW} = 4.5A	●		90	140	mA
Input Supply Current (Note 6)		●		3.8	5.4	mA
Shutdown Supply Current	$V_{\overline{SHDN}}$ = 0V, V_{SW} = 0V, V_C Open			15	50	μA
		●			75	μA
Lockout Threshold	V_C Open	●	2.3	2.38	2.46	V
Shutdown Thresholds	V_C Open Device Shutting Down	●	0.13	0.37	0.60	V
	Device Starting Up	●	0.25	0.45	0.7	V
Synchronization Threshold		●		1.5	2.2	V
Synchronizing Range			580		1000	kHz
SYNC Pin Input Resistance				40		kΩ

Note 1: Absolute Maximum Ratings are those values beyond which the life of a device may be impaired.

Note 2: Gain is measured with a V_C swing equal to 200mV above the switching threshold level to 200mV below the upper clamp level.

Note 3: Minimum input voltage is not measured directly, but is guaranteed by other tests. It is defined as the voltage where internal bias lines are still regulated so that the reference voltage and oscillator frequency remain constant. Actual minimum input voltage to maintain a regulated output will depend on output voltage and load current. See Applications Information.

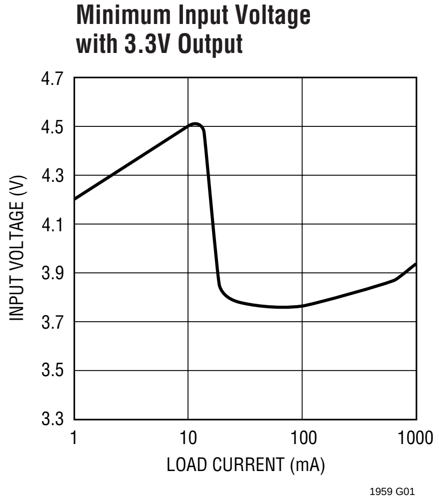
Note 4: This is the minimum voltage across the boost capacitor needed to guarantee full saturation of the internal power switch.

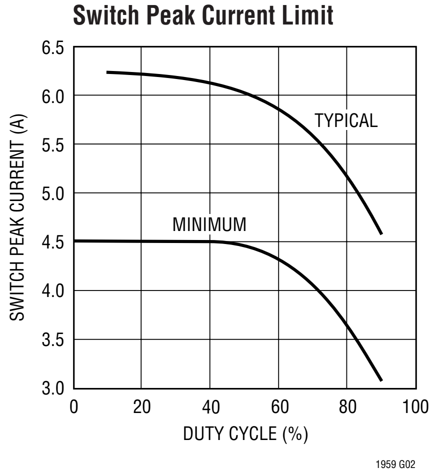
Note 5: Boost current is the current flowing into the boost pin with the pin held 5V above input voltage. It flows only during switch on time.

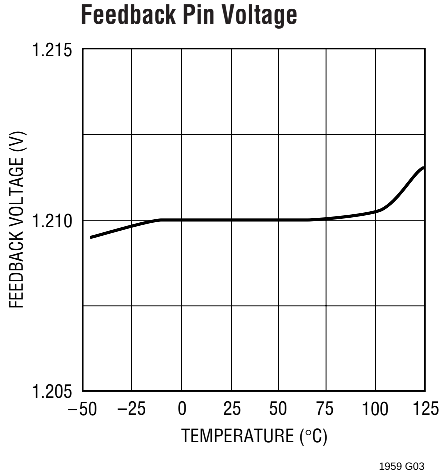
Note 6: Input supply current is the bias current drawn by the input pin with switching disabled.

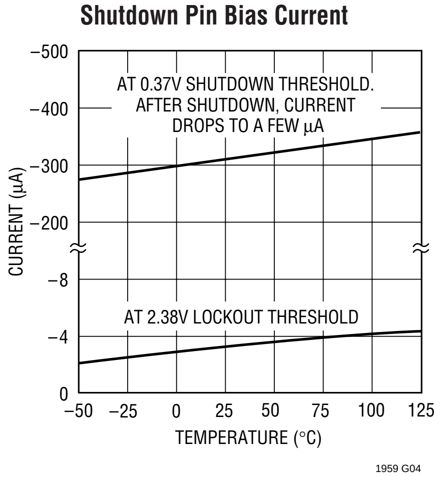
Note 7: Switch on resistance is calculated by dividing V_{IN} to V_{SW} voltage by the forced current (4.5A). See Typical Performance Characteristics for the graph of switch voltage at other currents.

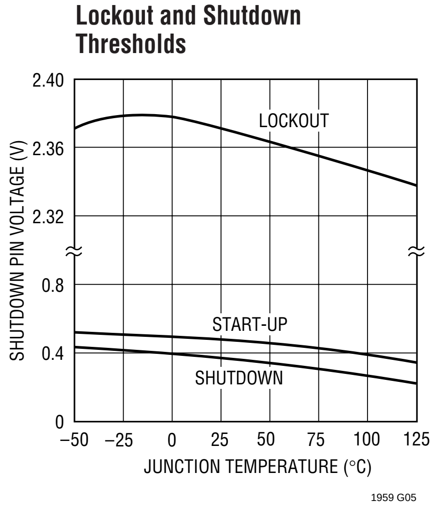
TYPICAL PERFORMANCE CHARACTERISTICS

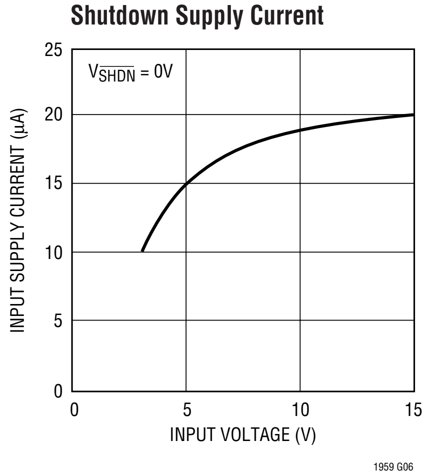


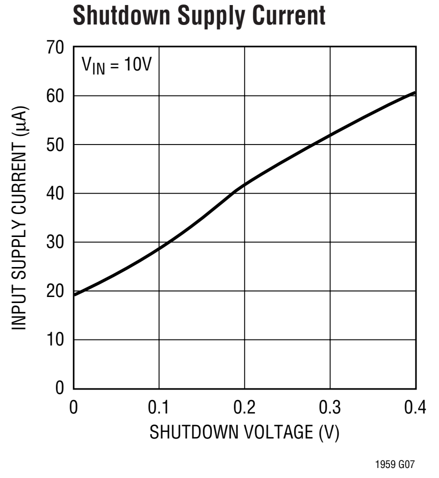


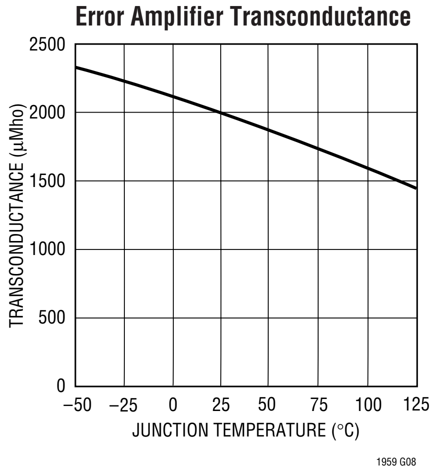


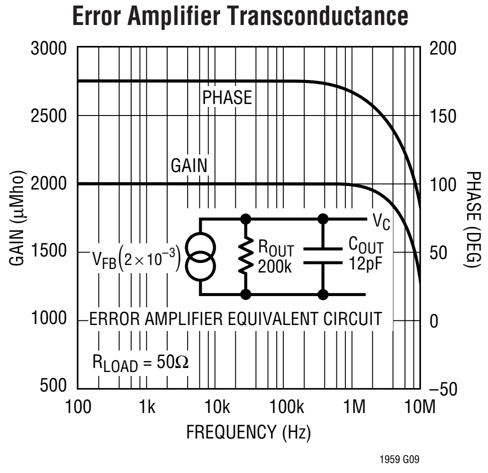






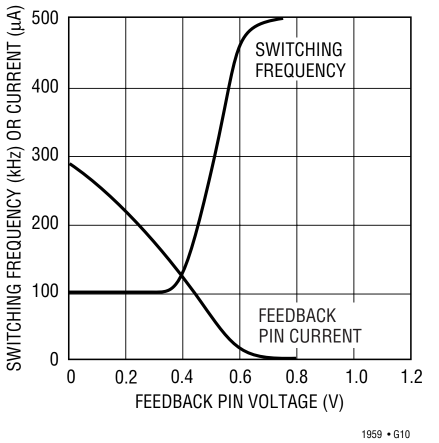




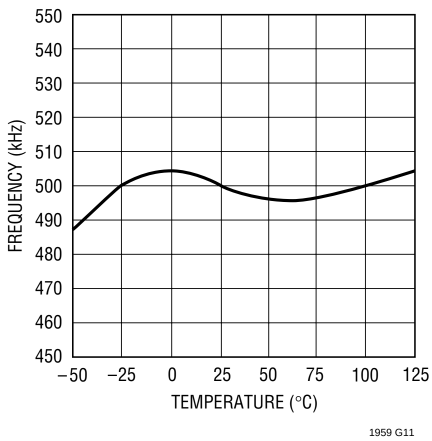


TYPICAL PERFORMANCE CHARACTERISTICS

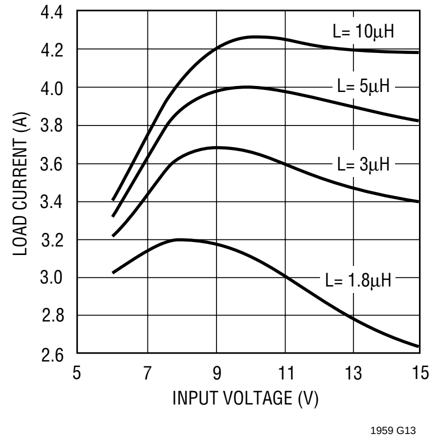
Frequency Foldback



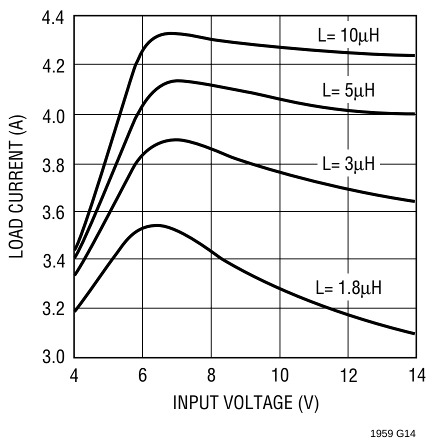
Switching Frequency



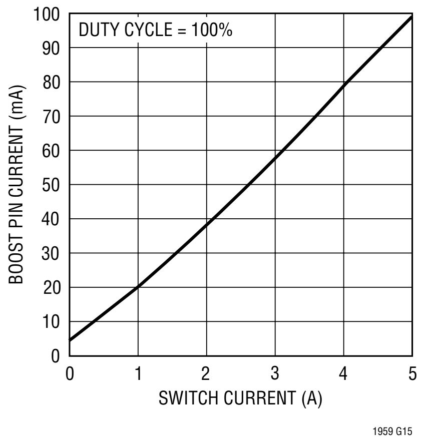
Maximum Load Current at V_{OUT} = 5V



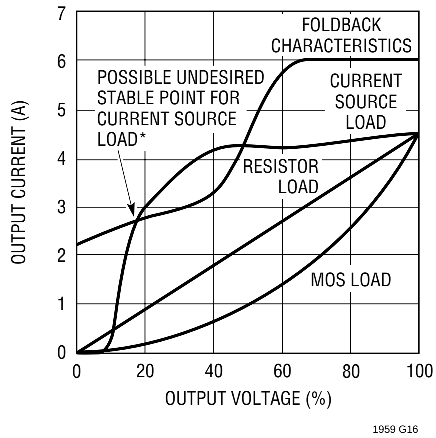
Maximum Load Current at V_{OUT} = 3.3V



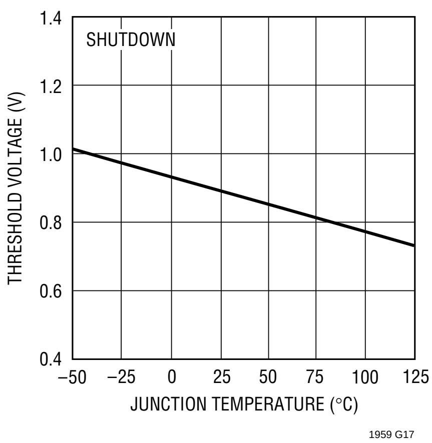
BOOST Pin Current



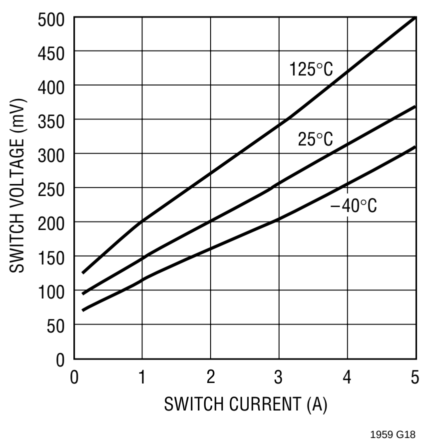
Current Limit Foldback



V_C Pin Shutdown Threshold



Switch Voltage Drop



*See "More Than Just Voltage Feedback" in the Applications Information section.

PIN FUNCTIONS

FB: The feedback pin is used to set output voltage using an external voltage divider that generates 1.21V at the pin with the desired output voltage. Three additional functions are performed by the FB pin. When the pin voltage drops below 0.8V, switch current limit is reduced. Below 0.7V the external sync function is disabled and switching frequency is reduced. See Feedback Pin Function section in Applications Information for details.

BOOST: The BOOST pin is used to provide a drive voltage, higher than the input voltage, to the internal bipolar NPN power switch. Without this added voltage, the typical switch voltage loss would be about 1.5V. The additional boost voltage allows the switch to saturate and voltage loss approximates that of a 0.07Ω FET structure, but with much smaller die area. Efficiency improves from 75% for conventional bipolar designs to > 89% for these new parts.

V_{IN}: This is the collector of the on-chip power NPN switch. This pin powers the internal circuitry and internal regulator. At NPN switch on and off, high dI/dt edges occur on this pin. Keep the external bypass and catch diode close to this pin. All trace inductance on this path will create a voltage spike at switch off, adding to the V_{CE} voltage across the internal NPN.

GND: The GND pin connection needs consideration for two reasons. First, it acts as the reference for the regulated output, so load regulation will suffer if the "ground" end of the load is not at the same voltage as the GND pin of the IC. This condition will occur when load current or other currents flow through metal paths between the GND pin and the load ground point. Keep the ground path short between the GND pin and the load and use a ground plane when possible. The second consideration is EMI caused by GND pin current spikes. Internal capacitance between the V_{SW} pin and the GND pin creates very narrow (<10ns) current spikes in the GND pin. If the GND pin is connected to system ground with a long metal trace, this trace may radiate excess EMI. Keep the path between the input bypass and the GND pin short. The GND pin of the SO-8 package is directly attached to the internal tab. This pin should be attached to a large copper area to improve thermal resistance.

V_{SW}: The switch pin is the emitter of the on-chip power NPN switch. This pin is driven up to the input pin voltage during switch on time. Inductor current drives the switch pin negative during switch off time. Negative voltage is clamped with the external catch diode. Maximum negative switch voltage allowed is –0.8V.

SYNC: (S08 Package Only) The sync pin is used to synchronize the internal oscillator to an external signal. It is directly logic compatible and can be driven with any signal between 10% and 90% duty cycle. The synchronizing range is equal to *initial* operating frequency, up to 1MHz. See Synchronizing section in Applications Information for details. When not in use, this pin should be grounded.

$\overline{\text{SHDN}}$: The shutdown pin is used to turn off the regulator and to reduce input drain current to a few microamperes. Actually, this pin has two separate thresholds, one at 2.38V to disable switching, and a second at 0.4V to force complete micropower shutdown. The 2.38V threshold functions as an accurate undervoltage lockout (UVLO). This is sometimes used to prevent the regulator from operating until the input votlage has reached a predetermined level.

V_C: The V_C pin is the output of the error amplifier and the input of the peak switch current comparator. It is normally used for frequency compensation, but can do double duty as a current clamp or control loop override. This pin sits at about 1V for very light loads and 2V at maximum load. It can be driven to ground to shut off the regulator, but if driven high, current must be limited to 4mA.

BLOCK DIAGRAM

The LT1959 is a constant frequency, current mode buck converter. This means that there is an internal clock and two feedback loops that control the duty cycle of the power switch. In addition to the normal error amplifier, there is a current sense amplifier that monitors switch current on a cycle-by-cycle basis. A switch cycle starts with an oscillator pulse which sets the R_S flip-flop to turn the switch on. When switch current reaches a level set by the inverting input of the comparator, the flip-flop is reset and the switch turns off. Output voltage control is obtained by using the output of the error amplifier to set the switch current trip point. This technique means that the error amplifier commands current to be delivered to the output rather than voltage. A voltage fed system will have low phase shift up to the resonant frequency of the inductor and output capacitor, then an abrupt 180° shift will occur. The current fed system will have 90° phase shift at a much lower frequency, but will not have the additional 90° shift until well beyond the LC resonant frequency. This makes it much easier to frequency compensate the feedback loop and also gives much quicker transient response.

High switch efficiency is attained by using the BOOST pin to provide a voltage to the switch driver which is higher than the input voltage, allowing switch to be saturated. This boosted voltage is generated with an external capacitor and diode. Two comparators are connected to the shutdown pin. One has a 2.38V threshold for undervoltage lockout and the second has a 0.4V threshold for complete shutdown.

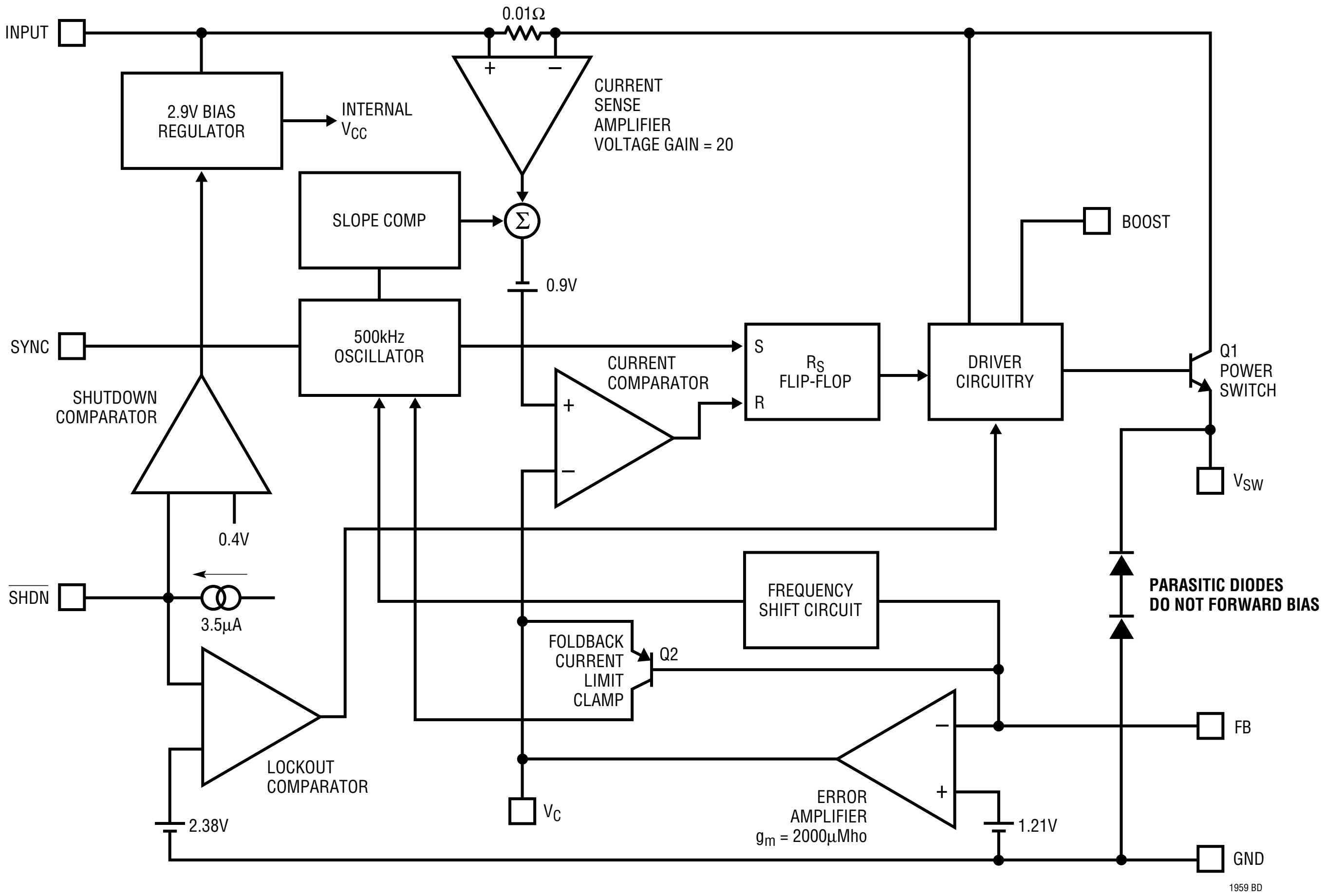


Figure 1. Block Diagram

APPLICATIONS INFORMATION

FEEDBACK PIN FUNCTIONS

The feedback (FB) pin on the LT1959 is used to set output voltage and provide several overload protection features. The first part of this section deals with selecting resistors to set output voltage and the remaining part talks about foldback frequency and current limiting created by the FB pin. Please read both parts before committing to a final design.

The suggested value for the output divider resistor (see Figure 2) from FB to ground (R2) is 2.5k or less, and a formula for R1 is shown below. The output voltage error caused by ignoring the input bias current on the FB pin is less than 0.1% with R2 = 2.5k. Please read the following if divider resistors are increased above the suggested values.

$$R1 = \frac{R2\left(V_{OUT} - 1.21\right)}{1.21}$$

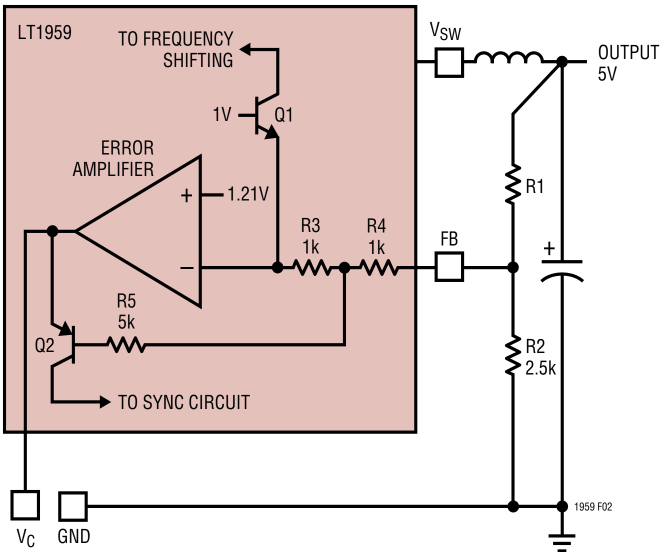


Figure 2. Frequency and Current Limit Foldback

More Than Just Voltage Feedback

The feedback pin is used for more than just output voltage sensing. It also reduces switching frequency and current limit when output voltage is very low (see the Frequency Foldback graph in Typical Performance Characteristics). This is done to control power dissipation in both the IC and in the external diode and inductor during short-circuit conditions. A shorted output requires the switching regulator to operate at very low duty cycles, and the average current through the diode and inductor is equal to the short-circuit current limit of the switch (typically 6A for the LT1959, folding back to less than 3A). Minimum switch on time limitations would prevent the switcher from attaining a sufficiently low duty cycle if switching frequency were maintained at 500kHz, so frequency is reduced by about 5:1 when the feedback pin voltage drops below 0.5V (see Frequency Foldback graph). This does not affect operation with normal load conditions; one simply sees a gear shift in switching frequency during start-up as the output voltage rises.

In addition to lower switching frequency, the LT1959 also operates at lower switch current limit when the feedback pin voltage drops below 0.8V. Q2 in Figure 2 performs this function by clamping the V_C pin to a voltage less than its normal 2.1V upper clamp level. This *foldback current limit* greatly reduces power dissipation in the IC, diode and inductor during short-circuit conditions. External synchronization is also disabled to prevent interference with foldback operation. Again, it is nearly transparent to the user under normal load conditions. The only loads that may be affected are current source loads which maintain full load current with output voltage less than 50% of final value. In these rare situations the feedback pin can be clamped above 0.75V with an external diode to defeat foldback current limit. *Caution:* clamping the feedback pin means that frequency shifting will also be defeated, so a combination of high input voltage and dead shorted output may cause the LT1959 to lose control of current limit.

APPLICATIONS INFORMATION

The internal circuitry which forces reduced switching frequency also causes current to flow out of the feedback pin when output voltage is low. The equivalent circuitry is shown in Figure 2. Q1 is completely off during normal operation. If the FB pin falls below 0.7V, Q1 begins to conduct current and reduces frequency at the rate of approximately 2kHz/µA. To ensure adequate frequency foldback (under worst-case short-circuit conditions), the external divider Thevinin resistance must be low enough to pull 150µA out of the FB pin with 0.3V on the pin ($R_{DIV} \le 2k$). *The net result is that reductions in frequency and current limit are affected by output voltage divider impedance. Although divider impedance is not critical, caution should be used if resistors are increased beyond the suggested values and short-circuit conditions will occur with high input voltage*. High frequency pickup will increase and the protection accorded by frequency and current foldback will decrease.

MAXIMUM OUTPUT LOAD CURRENT

Maximum load current for a buck converter is limited by the maximum switch current rating (I_P) of the LT1959. This current rating is 4.5A up to 50% duty cycle (DC), decreasing to 3.7A at 80% duty cycle. This is shown graphically in Typical Performance Characteristics and as shown in the formula below:

I_P = 4.5A for DC ≤ 50%
$I_P = 3.21 + 5.95(DC) - 6.75(DC)^2$ for 50% < DC < 90%

DC = Duty cycle = V_{OUT}/V_{IN}

Example: with V_{OUT} = 5V, V_{IN} = 8V; DC = 5/8 = 0.625, and;

$$I_{SW(MAX)} = 3.21 + 5.95(0.625) - 6.75(0.625)^2 = 4.3A$$

Current rating decreases with duty cycle because the LT1959 has internal slope compensation to prevent current mode subharmonic switching. For more details, read Application Note 19. The LT1959 is a little unusual in this regard because it has nonlinear slope compensation which gives better compensation with less reduction in current limit.

Maximum load current would be equal to maximum switch current *for an infinitely large inductor*, but with finite inductor size, maximum load current is reduced by one-half peak-to-peak inductor current. The following formula assumes continuous mode operation, implying that the term on the right is less than one-half of I_P.

$$I_{OUT(MAX)} = \text{Continuous Mode} \quad I_P - \frac{(V_{OUT})(V_{IN} - V_{OUT})}{2(L)(f)(V_{IN})}$$

For the conditions above and L = 3.3µH,

$$I_{OUT(MAX)} = 4.3 - \frac{(5)(8-5)}{2\left(3.3 \bullet 10^{-6}\right)\left(500 \bullet 10^{3}\right)(8)}$$
$$= 4.3 - 0.57 = 3.73A$$

At V_{IN} = 15V, duty cycle is 33%, so I_P is just equal to a fixed 4.5A, and $I_{OUT(MAX)}$ is equal to:

$$4.5 - \frac{(5)(15-5)}{2\left(3.3 \bullet 10^{-6}\right)\left(500 \bullet 10^{3}\right)(15)}$$
$$= 4.5 - 1.01 = 3.49A$$

Note that there is less load current available at the higher input voltage because inductor ripple current increases. This is not always the case. Certain combinations of inductor value and input voltage range may yield lower available load current at the lowest input voltage due to reduced peak switch current at high duty cycles. If load current is close to the maximum available, please check maximum available current at both input voltage extremes. To calculate actual peak switch current with a given set of conditions, use:

$$I_{SW(PEAK)} = I_{OUT} + \frac{V_{OUT}(V_{IN} - V_{OUT})}{2(L)(f)(V_{IN})}$$

APPLICATIONS INFORMATION

CHOOSING THE INDUCTOR AND OUTPUT CAPACITOR

For most applications the output inductor will fall in the range of 3µH to 20µH. Lower values are chosen to reduce physical size of the inductor. Higher values allow more output current because they reduce peak current seen by the LT1959 switch, which has a 4.5A limit. Higher values also reduce output ripple voltage, and reduce core loss. Graphs in the Typical Performance Characteristics section show maximum output load current versus inductor size and input voltage.

When choosing an inductor you might have to consider maximum load current, core and copper losses, allowable component height, output voltage ripple, EMI, fault current in the inductor, saturation, and of course, cost. The following procedure is suggested as a way of handling these somewhat complicated and conflicting requirements.

1. Choose a value in microhenries from the graphs of maximum load current and core loss. Choosing a small inductor with lighter loads may result in discontinuous mode of operation, but the LT1959 is designed to work well in either mode. Keep in mind that lower core loss means higher cost, at least for closed core geometries like toroids. The core loss graphs show absolute loss for a 3.3V output, so actual percent losses must be calculated for each situation.

   Assume that the average inductor current is equal to load current and decide whether or not the inductor must withstand continuous fault conditions. If maximum load current is 0.5A, for instance, a 0.5A inductor may not survive a continuous 4.5A overload condition. Dead shorts will actually be more gentle on the inductor because the LT1959 has foldback current limiting.

2. Calculate peak inductor current at full load current to ensure that the inductor will not saturate. Peak current can be significantly higher than output current, especially with smaller inductors and lighter loads, so don't omit this step. Powdered iron cores are forgiving because they saturate softly, whereas ferrite cores saturate abruptly. Other core materials fall in between somewhere. The following formula assumes continuous mode of operation, but it errs only slightly on the high side for discontinuous mode, so it can be used for all conditions.

$$I_{PEAK} = I_{OUT} + \frac{V_{OUT}\left(V_{IN} - V_{OUT}\right)}{2\left(f\right)\left(L\right)\left(V_{IN}\right)}$$

   V_{IN} = Maximum input voltage
   f = Switching frequency, 500kHz

3. Decide if the design can tolerate an "open" core geometry like a rod or barrel, which have high magnetic field radiation, or whether it needs a closed core like a toroid to prevent EMI problems. One would not want an open core next to a magnetic storage media, for instance! This is a tough decision because the rods or barrels are temptingly cheap and small and there are no helpful guidelines to calculate when the magnetic field radiation will be a problem.

4. Start shopping for an inductor (see representative surface mount units in Table 2) which meets the requirements of core shape, peak current (to avoid saturation), average current (to limit heating), and fault current (if the inductor gets too hot, wire insulation will melt and cause turn-to-turn shorts). Keep in mind that all good things like high efficiency, low profile, and high temperature operation will increase cost, sometimes dramatically. Get a quote on the cheapest unit first to calibrate yourself on price, then ask for what you really want.

5. After making an initial choice, consider the secondary things like output voltage ripple, second sourcing, etc. Use the experts in the Linear Technology's applications department if you feel uncertain about the final choice. They have experience with a wide range of inductor types and can tell you about the latest developments in low profile, surface mounting, etc.

APPLICATIONS INFORMATION

Table 2

VENDOR/ PART NO.	VALUE (μH)	DC (Amps)	CORE TYPE	SERIES RESIS-TANCE(Ω)	CORE MATER-IAL	HEIGHT (mm)
Coiltronics						
CTX2-1	2	4.1	Tor	0.011	KMμ	4.2
CTX5-4	5	4.4	Tor	0.019	KMμ	6.4
CTX8-4	8	3.5	Tor	0.020	KMμ	6.4
CTX2-1P	2	3.4	Tor	0.014	52	4.2
CTX2-3P	2	4.6	Tor	0.012	52	4.8
CTX5-4P	5	3.3	Tor	0.027	52	6.4
Sumida						
CDRH125	10	4.0	SC	0.025	Fer	6
CDRH125	12	3.5	SC	0.027	Fer	6
CDRH125	15	3.3	SC	0.030	Fer	6
CDRH125	18	3.0	SC	0.034	Fer	6
Coilcraft						
DT3316-222	2.2	5	SC	0.035	Fer	5.1
DT3316-332	3.3	5	SC	0.040	Fer	5.1
DT3316-472	4.7	3	SC	0.045	Fer	5.1
Pulse						
PE-53650	4	4.8	Tor	0.017	52	9.1
PE-53651	5	5.4	Tor	0.018	52	9.1
PE-53652	9	5.5	Tor	0.022	52	10
PE-53653	16	5.1	Tor	0.032	52	10
Dale						
IHSM-4825	2.7	5.1	Open	0.034	Fer	5.6
IHSM-4825	4.7	4.0	Open	0.047	Fer	5.6
IHSM-5832	10	4.3	Open	0.053	Fer	7.1
IHSM-5832	15	3.5	Open	0.078	Fer	7.1
IHSM-7832	22	3.8	Open	0.054	Fer	7.1

Tor = Toroid
SC = Semiclosed geometry
Fer = Ferrite core material
52 = Type 52 powdered iron core material
KMμ = Kool Mμ®

Output Capacitor

The output capacitor is normally chosen by its Effective Series Resistance (ESR), because this is what determines output ripple voltage. At 500kHz, any polarized capacitor is essentially resistive. To get low ESR takes *volume*, so physically smaller capacitors have high ESR. The ESR range for typical LT1959 applications is 0.05Ω to 0.2Ω. A typical output capacitor is an AVX type TPS, 100μF at 10V, with a guaranteed ESR less than 0.1Ω. This is a "D" size surface mount solid tantalum capacitor. TPS capacitors are specially constructed and tested for low ESR, so they give the lowest ESR for a given volume. The value in microfarads is not particularly critical, and values from 22μF to greater than 500μF work well, but you cannot cheat mother nature on ESR. If you find a tiny 22μF solid tantalum capacitor, it will have high ESR, and output ripple voltage will be terrible. Table 3 shows some typical solid tantalum surface mount capacitors.

Kool Mμ is a registered trademark of Magnetics, Inc.

Table 3. Surface Mount Solid Tantalum Capacitor ESR and Ripple Current

E Case Size	ESR (Max., Ω)	Ripple Current (A)
AVX TPS, Sprague 593D	0.1 to 0.3	0.7 to 1.1
AVX TAJ	0.7 to 0.9	0.4
D Case Size		
AVX TPS, Sprague 593D	0.1 to 0.3	0.7 to 1.1
C Case Size		
AVX TPS	0.2 (typ)	0.5 (typ)

Many engineers have heard that solid tantalum capacitors are prone to failure if they undergo high surge currents. This is historically true, and type TPS capacitors are specially tested for surge capability, but surge ruggedness is not a critical issue with the *output* capacitor. Solid tantalum capacitors fail during very high *turn-on* surges, which do not occur at the output of regulators. High *discharge* surges, such as when the regulator output is dead shorted, do not harm the capacitors.

Unlike the input capacitor, RMS ripple current in the output capacitor is normally low enough that ripple current rating is not an issue. The current waveform is triangular with a typical value of 200mA$_{RMS}$. The formula to calculate this is:

Output Capacitor Ripple Current (RMS):

$$I_{RIPPLE(RMS)} = \frac{0.29\left(V_{OUT}\right)\left(V_{IN} - V_{OUT}\right)}{\left(L\right)\left(f\right)\left(V_{IN}\right)}$$

APPLICATIONS INFORMATION

Ceramic Capacitors

Higher value, lower cost ceramic capacitors are now available in smaller case sizes. These are ideal for input bypassing because of their high ripple rating and tolerance to turn-on surges. As output capacitors, caution must be used. Solid tantalum capacitor's ESR generates a loop "zero" at 5kHz to 50kHz that is beneficial in giving acceptable loop phase margin. Ceramic capacitors remain capacitive to beyond 300kHz and usually resonate with their ESL before ESR becomes effective. When using ceramic output capacitors, the loop compensation pole frequency must be reduced by a typical factor of 10.

OUTPUT RIPPLE VOLTAGE

Figure 3 shows a typical output ripple voltage waveform for the LT1959. Ripple voltage is determined by the high frequency impedance of the output capacitor, and ripple current through the inductor. Peak-to-peak ripple current through the inductor into the output capacitor is:

$$I_{P\text{-}P} = \frac{\left(V_{OUT}\right)\left(V_{IN} - V_{OUT}\right)}{\left(V_{IN}\right)\left(L\right)\left(f\right)}$$

For high frequency switchers, the sum of ripple current slew rates may also be relevant and can be calculated from:

$$\Sigma\frac{dI}{dt} = \frac{V_{IN}}{L}$$

Peak-to-peak output ripple voltage is the sum of a *triwave* created by peak-to-peak ripple current times ESR, and a *square* wave created by parasitic inductance (ESL) and ripple current slew rate. Capacitive reactance is assumed to be small compared to ESR or ESL.

$$V_{RIPPLE} = \left(I_{P\text{-}P}\right)\left(ESR\right) + \left(ESL\right)\Sigma\frac{dI}{dt}$$

Example: with $V_{IN} = 10V$, $V_{OUT} = 5V$, $L = 10\mu H$, $ESR = 0.1\Omega$, ESL = 10nH:

$$I_{P\text{-}P} = \frac{\left(5\right)\left(10 - 5\right)}{\left(10\right)\left(10 \bullet 10^{-6}\right)\left(500 \bullet 10^{3}\right)} = 0.5A$$

$$\Sigma\frac{dI}{dt} = \frac{10}{10 \bullet 10^{-6}} = 10^{6}$$

$$V_{RIPPLE} = \left(0.5A\right)\left(0.1\right) + \left(10 \bullet 10^{-9}\right)\left(10^{6}\right)$$

$$= 0.05 + 0.01 = 60mV_{P\text{-}P}$$

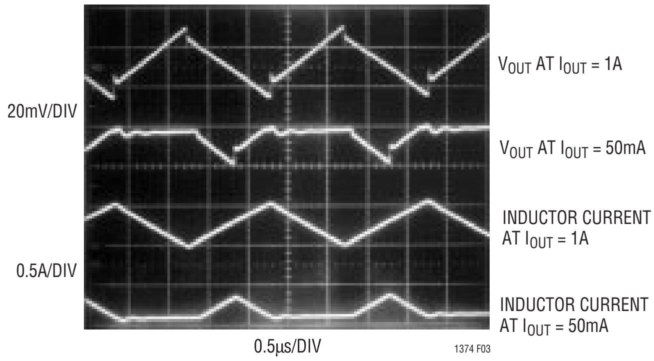


Figure 3. LT1959 Ripple Voltage Waveform

CATCH DIODE

The suggested catch diode (D1) is a 1N5821 Schottky, or its Motorola equivalent, MBR330. It is rated at 3A average forward current and 30V reverse voltage. Typical forward voltage is 0.5V at 3A. The diode conducts current only during switch off time. Peak reverse voltage is equal to regulator input voltage. Average forward current in normal operation can be calculated from:

$$I_{D(AVG)} = \frac{I_{OUT}\left(V_{IN} - V_{OUT}\right)}{V_{IN}}$$

This formula will not yield values higher than 3A with maximum load current of 4.25A unless the ratio of input to output voltage exceeds 3.4:1. The only reason to consider a larger diode is the worst-case condition of a high input voltage and *overloaded* (not shorted) output. Under short-circuit conditions, foldback current limit will reduce diode current to less than 2.6A, but if the output is overloaded

APPLICATIONS INFORMATION

and does not fall to less than 1/3 of nominal output voltage, foldback will not take effect. With the overloaded condition, output current will increase to a typical value of 5.7A, determined by peak switch current limit of 6A. With V_{IN} = 15V, V_{OUT} = 4V (5V overloaded) and I_{OUT} = 5.7A:

$$I_{D(AVG)} = \frac{5.7(15-4)}{15} = 4.18A$$

This is safe for short periods of time, but it would be prudent to check with the diode manufacturer if continuous operation under these conditions must be tolerated.

BOOST PIN CONSIDERATIONS

For most applications, the boost components are a 0.27μF capacitor and a 1N914 or 1N4148 diode. The anode is connected to the regulated output voltage and this generates a voltage across the boost capacitor nearly identical to the regulated output. In certain applications, the anode may instead be connected to the unregulated input voltage. This could be necessary if the regulated output voltage is very low (< 3V) or if the input voltage is less than 5V. Efficiency is not affected by the capacitor value, but the capacitor should have an ESR of less than 1Ω to ensure that it can be recharged fully under the worst-case condition of minimum input voltage. Almost any type of film or ceramic capacitor will work fine.

For nearly all applications, a 0.27μF boost capacitor works just fine, but for the curious, more details are provided here. The size of the boost capacitor is determined by switch drive current requirements. During switch on time, drain current on the capacitor is approximately I_{OUT}/50. At peak load current of 4.25A, this gives a total drain of 85mA. Capacitor ripple voltage is equal to the product of on time and drain current divided by capacitor value; $\Delta V = (t_{ON})(85mA/C)$. To keep capacitor ripple voltage to less than 0.6V (a slightly arbitrary number) at the worst-case condition of t_{ON} = 1.8μs, the capacitor needs to be 0.27μF. Boost capacitor ripple voltage is not a critical parameter, but if the minimum voltage across the capacitor drops to less than 3V, the power switch may not saturate fully and efficiency will drop. An *approximate* formula for absolute minimum capacitor value is:

$$C_{MIN} = \frac{(I_{OUT}/50)(V_{OUT}/V_{IN})}{(f)(V_{OUT}-2.8V)}$$

f = Switching frequency
V_{OUT} = Regulated output voltage
V_{IN} = Minimum input voltage

This formula can yield capacitor values substantially less than 0.27μF, but it should be used with caution since it does not take into account secondary factors such as capacitor series resistance, capacitance shift with temperature and output overload.

SHUTDOWN FUNCTION AND UNDERVOLTAGE LOCKOUT

Figure 4 shows how to add undervoltage lockout (UVLO) to the LT1959. Typically, ULVO is used in situations where the input supply is *current limited*, or has a relatively high source resistance. A switching regulator draws constant power from the source, so source current increases as source voltage drops. This looks like a negative resistance load to the source and can cause the source to current limit or latch low under low source voltage conditions. ULVO prevents the regulator from operating at source voltages where these problems might occur.

Threshold voltage for lockout is about 2.38V. A 3.5μA bias current flows *out* of the pin at threshold. This internally generated current is used to force a default high state on the shutdown pin if the pin is left open. When low shutdown current is not an issue, the error due to this current can be minimized by making R_{LO} 10k or less. If shutdown current is an issue, R_{LO} can be raised to 100k, but the error due to initial bias current and changes with temperature should be considered.

$$R_{LO} = 10k \text{ to } 100k \text{ (25k suggested)}$$

$$R_{HI} = \frac{R_{LO}(V_{IN}-2.38V)}{2.38V-R_{LO}(3.5\mu A)}$$

V_{IN} = Minimum input voltage